# 2nd Order DEs: Damped Motion

Questions 1–24

**Question 1.** A particle of mass 2 kg moves along a horizontal line. At time $t$, its displacement from a fixed point $O$ is $x$ metres, measured positive to the right. While $P$ moves, two horizontal forces act on it: a restoring force of magnitude $10x$ N acting towards $O$ and a resistance force of magnitude $6v$ N, where $v$ m s$^{-1}$ is the speed of $P$.

a) Write down the restoring force in terms of $x$, including its sign.

b) Write down the resistance force in terms of $\dfrac{dx}{dt}$, including its sign.

c) Show that the equation of motion is $\dfrac{d^2x}{dt^2} + 3\dfrac{dx}{dt} + 5x = 0$.

**Question 2.** A particle of mass 0.5 kg moves in a straight line. Its displacement from a fixed point $A$ is $x$ metres, measured positive away from $A$. At time $t$, the particle is moving with velocity $v$ m s$^{-1}$. The forces acting on the particle include a force of magnitude $4x$ N acting towards $A$, and a resistance force of magnitude $2v$ N. Show that the equation of motion is

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 8x = 0.$$

**Question 3.** A particle $P$ moves in a straight line. Its displacement from a fixed point $O$ is $x$ metres. The particle has mass $m$ kg. Two forces act on $P$, including a force of magnitude $kx$ N directed towards $O$, and a resistance force of magnitude $\lambda v$ N. Show that the equation of motion is

$$m\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = 0.$$

**Question 4.** A particle of mass 3 kg moves on the $x$-axis. Its displacement from the origin is $x$ metres. At time $t$, three horizontal forces act on the particle, including: a restoring force of magnitude $12x$ N, a resistance force of magnitude $9v$ N, and a driving force of magnitude $15\cos t$ N acting in the positive $x$-direction. Show that the equation of motion is

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 4x = 5\cos t.$$

**Question 5.** A particle $P$ of mass 1 kg moves in a horizontal straight line. Its displacement from a fixed point $O$ is $x$ metres. A restoring force of magnitude $8x$ N acts on $P$ towards $O$, and a resistance force of magnitude $4v$ N acts on $P$. Initially, the particle is at rest 2 metres from $O$.

a) Show that $\dfrac{d^2x}{dt^2} + 4\dfrac{dx}{dt} + 8x = 0$.

b) Find $x$ as a function of $t$.

**Question 6.** A particle of mass 2 kg moves on the $x$-axis. Its displacement from the origin is $x$ m. A restoring force of magnitude $6x$ N acts towards the origin, and a resistive force of magnitude $8v$ N acts on the particle. Initially, the particle is 3 metres from the origin and is travelling at 1 m $s^{-1}$ towards the origin. Find $x$ as a function of $t$.

**Question 7.** A particle $P$ of mass 0.8 kg moves in a horizontal straight line. At time $t$ seconds, its displacement from a fixed point $O$ is $x$ metres and its velocity is $v$ m s$^{-1}$. A force of magnitude $5x$ N acts on $P$ in the direction $PO$. The particle is also subject to a resistance force of magnitude $2v$ N. Initially, the particle is 1.2 metres from the origin and is travelling at 3 m s$^{-1}$ in the direction of $OP$.

a) Show that the motion satisfies $4\dfrac{d^2x}{dt^2} + 10\dfrac{dx}{dt} + 25x = 0$.

b) Find the position of $P$ when $t = 1$.

**Question 8.** A particle $P$ of mass 1.5 kg moves along the $x$-axis with displacement from the origin is $x$ m. Three forces act on $P$: a restoring force of magnitude $7.5x$ N towards the origin, a resistance force of magnitude $6v$ N, and a driving force $9\sin 2t$ N acting in the positive $x$-direction. Initially, $x = 4$ and the particle is travelling at 1 m s$^{-1}$ away from the origin.

a) Show that the motion satisfies $\dfrac{d^2x}{dt^2} + 4\dfrac{dx}{dt} + 5x = 6\sin 2t$.

b) Find $x$ as a function of $t$.

c) Describe the motion of $P$ for large values of $t$.

**Question 9.** A mass attached to a vertical spring is released from rest after being displaced from its equilibrium position. The displacement $x$ m from equilibrium at time $t$ s satisfies

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0.$$

Initially, the mass is pulled down 0.08 m and released from rest.

a) Find the displacement $x$ as a function of $t$.
b) State whether the motion is oscillatory and justify your answer.
c) Determine the time of the first maximum displacement after release.

**Question 10.** The angular displacement $\theta$ radians of a hinged panel from its equilibrium position is modelled by

$$\frac{d^2\theta}{dt^2} + 2\frac{d\theta}{dt} + 10\theta = 0.$$

The panel is initially released from rest after being displaced 0.2 radians.

a) Find $\theta$ in terms of $t$.
b) State the type of damping in the system.
c) Describe the motion as $t \to \infty$.

**Question 11.** A particle of mass $m$ kg is attached to the lower end of a light vertical spring. The particle moves only in the vertical direction. Its displacement from equilibrium at time $t$ is $x$ metres, measured downwards. While the particle is moving, two forces act on it in addition to its weight: a restoring force of magnitude $18x$ N acting towards the equilibrium position, and a resistive force of magnitude $\lambda v$ N, where $v$ m s$^{-1}$ is the speed of the particle. The particle is pulled down slightly and released.

a) Show that the motion is modelled by $m\dfrac{d^2x}{dt^2} + \lambda\dfrac{dx}{dt} + 18x = 0$.

b) Write down the auxiliary equation corresponding to the differential equation.

c) It is given that $m = 2$ and $\lambda = 8$.

i) Find the general solution for $x$.

ii) State whether the motion is oscillatory.

d) Keeping $m = 2$, determine the value of $\lambda$ for which the motion is critically damped.

e) Using your value of $\lambda$ from part (d), describe the motion for $m = 3$.

**Question 12.** A particle $M$ of mass 2 kg is connected to a point $C$ by a light elastic spring $CD$. The system lies on a smooth horizontal surface and is initially at rest with the spring at its natural length. At time $t = 0$, the point $D$ is set in motion and moves away from $C$ in a straight line with constant speed $U$. As the particle moves it experiences a resistive force proportional to its velocity. Let $x$ denote the extension of the spring at time $t$. The motion of the system is modelled by the differential equation

$$2\frac{d^2x}{dt^2} + 7k\frac{dx}{dt} + 6k^2x = 4kU.$$

Find $x$ in terms of $t$, $k$ and $U$.

**Question 13.** A mass is attached to a vertical spring with a damper. The displacement of the mass from its equilibrium position is $x$ cm at time $t$ seconds after the motion begins. Upward displacement is taken as positive. The motion is modelled by the differential equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 4x = 10e^{-2t}.$$

In one experiment, the mass is initially positioned 18 cm above its equilibrium position and is given an initial velocity of 8 cm $s^{-1}$ upwards.

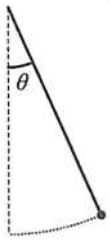
a) Find an expression for $x$ in terms of $t$.
b) Find, to the nearest mm, the maximum displacement of the mass above its equilibrium position.
c) Determine whether the model predicts that the mass ever returns to its equilibrium position. Justify your answer.

**Question 14.** A particle of mass 1 kg is attached to a spring and moves along a horizontal line. Its displacement from equilibrium is $x$ m at time $t$ s. The particle experiences a resistive force proportional to its velocity and a small periodic driving force. The motion is modelled by

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 13x = 10\cos 2t.$$

a) Find an expression for $x$ in terms of $t$.
b) Hence find the steady-state solution.
c) Find the amplitude of the steady-state solution.

**Question 15.** The angular displacement of a pendulum from the downward vertical is $\theta$ radians at time $t$ seconds, as shown in the diagram below.




When air resistance and a small periodic driving force are taken into account, the motion of the pendulum is modelled by

$$\frac{d^2\theta}{dt^2} + 8\frac{d\theta}{dt} + 32\theta = 80\cos 4t.$$

a) Find the general solution of the differential equation.

Initially the pendulum is at the equilibrium position and is given an angular velocity of 9 rad $s^{-1}$ in the positive direction.

b) Find an expression for $\theta$ in terms of $t$.

c) Describe the motion of the pendulum as $t$ becomes large.

**Question 16.** A pendulum swings in a vertical plane about a fixed point. The angular displacement from the downward vertical is $\theta$ radians at time $t$ seconds. Air resistance acts on the pendulum and a small magnetic impulse is applied when the motion begins. The motion is modelled by

$$\frac{d^2\theta}{dt^2} + 6\frac{d\theta}{dt} + 9\theta = 9e^{-t}.$$

Initially the pendulum is at the downward vertical and at rest. Show that, according to the model, the pendulum never returns to the downward vertical for any $t > 0$.

**Question 17.** A heavy door in a warehouse is fitted with a spring-loaded closing mechanism. When the door reaches the frame, it compresses a horizontal spring which slows the door and eventually brings it to rest. The door and its fittings have total mass $M$ kg. When the spring is compressed by $y$ metres, it exerts a force of magnitude $M\alpha^2 y$ N on the door. Let $y$ metres be the compression of the spring $t$ seconds after the door first makes contact with it.

a) Assuming no other horizontal forces act on the door, show that while the spring is being compressed the motion satisfies $\dfrac{d^2y}{dt^2} + \alpha^2 y = 0$.

b) The door first touches the spring with speed $W$. Write down expressions for $y$ and $\dfrac{dy}{dt}$ when $t = 0$.

c) Find the compression of the spring as a function of time.

d) Find the maximum compression of the spring.

In reality, the closing mechanism contains oil which produces a resistive force proportional to the speed of the door. The magnitude of this force is $\lambda M v$ N, where $v$ is the speed of the door and $\lambda$ is a positive constant.

e) Show that the compression now satisfies $\dfrac{d^2y}{dt^2} + \lambda\dfrac{dy}{dt} + \alpha^2 y = 0$.

f) When $\lambda = 3\alpha$, determine whether the motion is Heavy Damped (overdamped), Critically Damped, or Lightly Damped (underdamped). Justify your answer.

g) When $\lambda = 3\alpha$, describe the motion of the door after it first touches the spring.

**Question 18.** A heavy door is fitted with a spring-loaded hinge and a hydraulic damping mechanism. When displaced from equilibrium, the hinge exerts a restoring moment proportional to $\theta$, and the damping mechanism provides a resistive moment proportional to $\dfrac{d\theta}{dt}$. The angle $\theta$ radians through which the door is open at time $t$ seconds is modelled by

$$\frac{d^2\theta}{dt^2} + 4\frac{d\theta}{dt} + 13\theta = 26.$$

Initially, the door is opened to an angle of 3 radians and released from rest.

a) Find the particular solution for $\theta$.

b) Show that the solution may be written in the form $\theta = 2 + Re^{-2t}\cos(3t - \alpha)$ where $R$ and $\alpha$ are constants.

c) State the limiting value of $\theta$ as $t \to \infty$, and explain its physical meaning.

d) Sketch the graph of $x$ against $t$, showing clearly the equilibrium position.

**Question 19.** A crate of mass $M$ kg is suspended from an elastic cable attached to a winch above it. The crate is allowed to drop vertically, and the cable first becomes taut at time $t = 0$. For $t$ seconds after the cable first becomes taut, the extension of the cable is $y$ metres, and the speed of the crate is $v$ m s$^{-1}$. A model for the motion while the cable remains taut assumes that the forces acting on the crate are its weight, a tension in the cable of magnitude $Cy$ N, and a resistive force of magnitude $Dv$ N. Here $C$ and $D$ are positive constants, and it is given that $4CM > D^2$. Take downwards as the positive direction.

a) Show that the motion is modelled by the differential equation $M\dfrac{d^2y}{dt^2} + D\dfrac{dy}{dt} + Cy = Mg$.

b) Hence show that the general solution may be written in the form

$$y = e^{-\frac{D}{2M}t}\left(P\cos\left(\frac{\sqrt{4CM - D^2}}{2M}t\right) + Q\sin\left(\frac{\sqrt{4CM - D^2}}{2M}t\right)\right) + \frac{Mg}{C},$$

where $P$ and $Q$ are constants.

It is now given that $C = 25$, $D = 24$, $M = 9$, and $g = 10$, and that when the cable first becomes taut the extension is zero and the crate is moving downwards with speed 9 m s$^{-1}$.

c) Find the values of $P$ and $Q$, and hence write down the corresponding expression for $y$ in terms of $t$.

d) Show that the crate reaches its greatest extension when $\tan t = \frac{9}{2}$ and hence find the time at which the greatest extension occurs.

e) State the limiting value of $y$ as $t \to \infty$, and explain what this means in the context of the motion.

A second model ignores the resistive force completely. The values of $C$, $M$ and $g$ remain unchanged.

f) Find the corresponding expression for $y$ in terms of $t$.

g) Use the solutions from parts (c), (d) and (f) to show that the greatest extension predicted by the undamped model is larger than the greatest extension predicted by the damped model.

h) Sketch the expressions for $y$ in terms of $t$ for both the damped model and the undamped model on the same axes, identifying clearly which is which.

**Question 20. Prerequisite knowledge for Q20: Mean Value of a Function**

A particle $Q$ moves along the $x$-axis. At time $t$ seconds, its displacement from the origin $O$ is $x$ metres and its velocity is $v$ m s$^{-1}$. The motion of $Q$ is modelled by the differential equation $\ddot{x} + \lambda\dot{x} = 0$, where $\lambda$ is a positive constant. Initially $Q$ is projected from $O$ with speed $u$ in the positive $x$-direction.

a) Using $v = \dfrac{dx}{dt}$, show that the motion satisfies $\dfrac{dv}{dx} + \lambda = 0$.

b) Hence show that while the particle is moving, $v = u - \lambda x$.

The particle first comes to instantaneous rest at the point $A$, where $x = d$.

c) Show that $u = \lambda d$.

d) Hence prove that $v = \lambda(d - x)$.

e) The quantity $z$ is defined by $z = \dfrac{1}{v}$. Show that the mean value of $z$ with respect to the displacement as the particle moves from $O$ to the point where $x = \frac{1}{2}d$ can be written as $\dfrac{\ln 4}{u}$.

**Question 21.** The motion of a particle is given by

$$x = 5e^{-0.2t}\cos 3t.$$

a) Find the amplitude of the motion at time $t$.
b) Find the ratio of the amplitudes at time $t$ and $t + T$, where $T$ is the period of the motion.

**Question 22.** A particle oscillates vertically about an equilibrium position. Its displacement from equilibrium is $y$ cm at time $t$ seconds. A model for the motion of the particle is

$$\frac{d^2y}{dt^2} + 2c\frac{dy}{dt} + 10y = 0,$$

where $c$ is a positive constant.

a) Given that the motion is oscillatory, find the general solution.

It is observed experimentally that the amplitude of successive oscillations decreases so that the amplitude after each complete oscillation is 0.9 times the previous amplitude.

b) Show that $e^{-cT} = 0.9$, where $T$ is the period of the oscillations.
c) Hence determine the value of $c$.

Initially the particle passes through equilibrium moving in the positive direction with speed 15 m $s^{-1}$.

d) Find an expression for $y(t)$, writing constants to three significant figures.

**Question 23.** A glider $G$ of mass $m$ kg moves along a straight horizontal air track through a fixed equilibrium point $E$. At time $t$ seconds, the displacement of $G$ from $E$ is $x$ metres and its velocity is $v$ m s$^{-1}$. When $G$ is displaced, the track's control mechanism exerts a restoring force of magnitude $4mx$ N directed towards $E$. Initially, the glider is 1 m to the right of $E$ and is moving further to the right with speed 2 m s$^{-1}$.

a) Show that, while only the restoring force acts, the motion satisfies $\dfrac{d^2x}{dt^2} + 4x = 0$.

b) Find $x$ in terms of $t$.

c) Find the amplitude and period of the motion.

d) Name the type of motion.

A resistive force of magnitude $2mv$ N is now introduced.

e) Show that the equation of motion becomes $\dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + 4x = 0$.

f) Determine the general solution of this differential equation.

g) State, with a reason, whether the motion is heavy damped (overdamped), critically damped, or light damped (underdamped).

Finally, while the resistive force is still present, a driving force of magnitude $4m\cos t$ N acts on the glider, so that the motion is modelled by

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 4x = 4\cos t.$$

At $t = 0$, the glider is again 1 m to the right of $E$ and has velocity 2 m s$^{-1}$ to the right.

h) Find $x(t)$.

i) Show that, for large values of $t$, the glider performs simple harmonic motion with period just over 6 seconds.

j) Find the amplitude of this long-term motion.

k) Show that the part of the solution in part (h) multiplied by $e^{-t}$ can be written in the form $Ce^{-t}\cos(\sqrt{3}t - \beta)$ where $C > 0$.

l) Hence show that $x_s(t) - Ce^{-t} \le x(t) \le x_s(t) + Ce^{-t}$ where $x_s(t) = \frac{12}{13}\cos t + \frac{8}{13}\sin t$.

m) Sketch the graph of $x$ against $t$, showing these bounding curves.

**Question 24.** A light horizontal spring is attached to a particle and a damping device. The particle is also acted on by a periodic driving force. Its displacement $x$ m from equilibrium at time $t$ s is found to be

$$x = e^{-\frac{1}{2}t}(A\cos 2t + B\sin 2t) + \cos t - \sin t,$$

where $A$ and $B$ are constants.

a) Show that $x$ satisfies a differential equation of the form

$$p\frac{d^2x}{dt^2} + q\frac{dx}{dt} + rx = m\cos t + n\sin t,$$

where $p$, $q$, $r$, $m$ and $n$ are integers to be found.

b) Sketch a graph of $x$ against $t$ in the case where $A = 1$ and $B = 1$.